

Top margin: 1inch
For all pages

Left margin: 1.5inch
For all pages

Right margin: 1inch
For all pages

A TECHNICAL REPORT ON THE STUDENT INDUSTRIAL WORK

EXPERIENCE SCHEME (SIWES II)

UNDERTAKEN AT

EMERALD NIGERIA LIMITED

2 ABUJA STREET, ENUGU CITY, LAGOS, NIGERIA.

Front page font:
Times New Roman,
14pt, all caps

BY

CIROMA, CHUKWUMA ADEKUNLE

CHE/1999/2000

SUBMITTED TO

THE SIWES COORDINATOR

DEPARTMENT OF CHEMICAL ENGINEERING

FACULTY OF TECHNOLOGY, OBAFEMI AWOLOWO

UNIVERSITY, ILE-IFE, OSUN STATE.

NOVEMBER 2002

Bottom margin: 1inch
For all pages

Department of Chemical Engineering,
Obafemi Awolowo University,
Faculty of Technology,
Ile-Ife, Osun State.
November 11th, 2002.

The SIWES coordinator,
Department of Chemical Engineering,
Obafemi Awolowo University,
Ile-Ife, Osun State.

Dear Sir,

LETTER OF TRANSMITTAL

I, CIROMA Chukwuma Adekunle with matriculation number CHE/1999/2000, hereby certify that I compiled this report based on my six-month Student Industrial Work Experience Scheme II (SIWES II) program undertaken at Emerald Nigeria Limited from April 2nd, 2002 to September 30th, 2002.

This report details the activities carried out during the period of the program. Thank you in anticipation of acknowledgement of the report submission.

Yours faithfully,

Your signature goes 
here

CIROMA Chukwuma Adekunle
CHE/1999/2000

CERTIFICATION

This is to certify that this report was compiled by CIROMA Chukwuma Adekunle with matriculation number CHE/1999/2000 on completion of the SIWES II program undertaken at Emerald Nigeria Limited, Lagos for the period of six months (April to September 2002).

 Your signature goes
here

.....

CIROMA Chukwuma Adekunle

Nov. 11th, 2002.

ACKNOWLEDGEMENT

Not more than 2 short paragraphs.

ABSTRACT

About 3 or 4 paragraphs that gives a summary of the project. Contains a brief introduction to the work, methodology, results and conclusion.

TABLE OF CONTENTS

LETTER OF TRANSMITTAL.....	ii
CERTIFICATION	iii
ACKNOWLEDGEMENT	iv
ABSTRACT	v
TABLE OF CONTENTS	vi
LIST OF TABLES	vii
LIST OF FIGURES	viii
NOMENCLATURE/ABBREVIATIONS	ix
CHAPTER ONE	1
INTRODUCTION	1
1.1 First order subheading	1
REFERENCES	5

LIST OF TABLES

Table 1: Material Streams

4

LIST OF FIGURES

Figure 1 - Gas-Liquid Separators	3
----------------------------------	---

NOMENCLATURE/ABBREVIATIONS

Definitions of nomenclature specific to your work as mentioned in the report or full meaning of abbreviations used in report.

CHAPTER ONE

Main headings should be 14pt, bold, Times New Roman

INTRODUCTION

1.1 First order subheading

First subheadings should be 12pt, bold, Times New Roman, aligned to the left

Block style paragraphs are usually preferred for technical writing.

All text body should be 12pt, double spaced, Times New Roman.

A technical report is a structured document that communicates detailed information about a specific technical or scientific activity, experiment, design, process, or research project. It serves as a formal medium through which engineers, scientists, and researchers share findings, methodologies, and recommendations with a target audience—often including supervisors, funding agencies, academic peers, or industrial stakeholders. Unlike general essays or journalistic writing, technical reports focus heavily on factual accuracy, clarity, and reproducibility. They are expected to present complex information in a way that can be understood, verified, and used by others working in the same or related fields (Gerson & Gerson, 2020).

Note how literature by two authors is cited in body of text. Both names are written.

A single line is left in between paragraphs.

Distillation remains one of the most important and widely used separation processes in the chemical and petrochemical industries. Its robustness and ability to handle large-scale separations make it indispensable, yet it is also among the most energy-intensive unit operations. Ongoing research has focused on improving column efficiency, reducing energy demand, and optimizing operating conditions through innovations such as structured packings, dividing-wall columns, and reactive distillation systems. These advancements have shown potential for enhancing mass transfer, minimizing pressure drop, and integrating multiple separation steps within a single column. By combining process intensification techniques with advanced modeling and simulation tools, modern distillation design is gradually evolving toward more sustainable and cost-effective operation (Kenig et al., 2001).

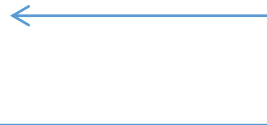
Note how literature by more than two authors is cited in body of text. Only the name of the first author is written with et al. showing the presence of other authors.

Notice how the pages of the main report are numbered.

Distillation efficiency depends strongly on the thermodynamic behavior of the mixture and the hydrodynamic characteristics of the column. Understanding vapor–liquid equilibrium, mass transfer rates, and phase interaction mechanisms is essential for accurate design and scale-up. Over the years, researchers have developed rigorous mathematical models that describe the performance of both tray and packed columns under various operating conditions. These models enable engineers to predict composition profiles, energy consumption, and column behavior without relying solely on experimental data. Advances in computational tools have further enhanced the accuracy of such simulations, allowing for optimization of column design and operating parameters to achieve higher separation efficiency and lower energy demand (Kister, 1992).

← Note how literature by one author is cited in body of text.

1.1.1 Second order subheading



1.1.1 Second order subheading



Second order subheadings can take any of the two forms.
1. Indented, Bold, 12pt, Times New Roman.
2. Non-indented, Italics, Bold, 12pt, Times New Roman.

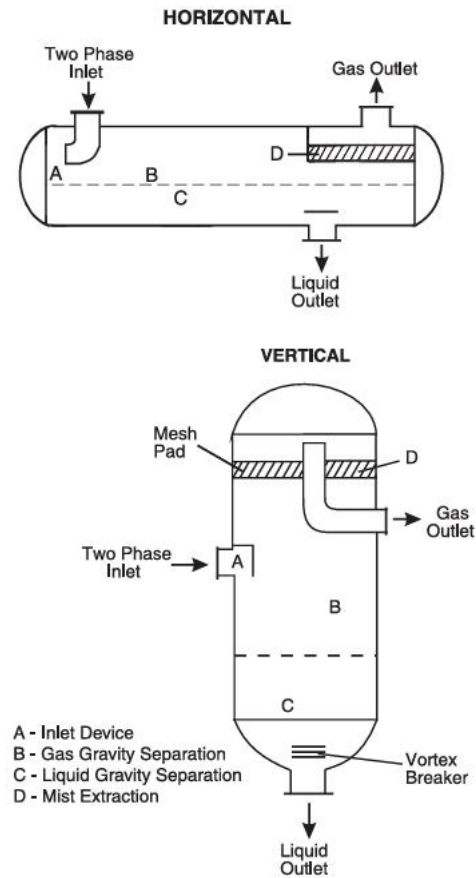


Figure 1 - Gas Liquid Separators

Captions should align to bottom left of diagram and be in 10pt, Times New Roman

Diagrams should be centralised and should have their own page i.e. no other text should be on the page the diagram is except the caption or in some cases, the source of the diagram (if external).

Also, the diagram should have been referred to in previous text. E.g. Figure 1 shows the schematic of gas-liquid separators.

Please, do not use spatial positions e.g 'figure 1 above/below' .

Table titles should align to top
left of table and be in 12pt, Times
New Roman

Table 1: Material Streams

Material Streams				
		Feed	Distillate	Bottoms
Vapour Fraction		0.0000	0.0000	0.0000
Temperature	C	60.0	82.02	110.7
Pressure	kPa	106.0	101.0	111.0
Molar Flow	kgmole/h	100.0	40.00	60.00
Mass Flow	kg/h	8653	3178	5475
Liquid Volume Flow	m ³ /h	9.896	3.608	6.288
Heat Flow	kJ/h	3.284e+006	2.165e+006	1.742e+006

Tables, as with figures, should have their dedicated
pages.

No internal borders on the table except an horizontal
line separating the column headings from the values as
shown in 1 and 2 above.

REFERENCES

Note
indentation
after first
line.



Akanbi, C. T. (1993). Drying and rehydration of skim milk powder: moisture, water activity and electron microscopy. *Nigerian Food Journal*. 11, 98-108. [Journal](#)

Leet, L.D. and Judson, S. (1965). *Physical Geology*, 3rd ed. Prentice Hall, London. [Book](#)

Oladebo, K. T. (1993). Flood frequency analysis by peaks-over-a-threshold method.

Unpublished M.Sc. Thesis, Department of Civil Engineering, University of

Dar Es Salaam, Tanzania. [Dissertation/Thesis \(Unpublished\)](#)

Ukponmwan, O. E. (1992). The mystery of sleep. In: Nigeria and the Challenge of

Knowledge. K. Adetugbo and A. E. Akingbohunbe, eds. Anchorprint Ltd.,

Ile-Ife. [Edited book \(containing articles written by different authors\)](#)

Note, in
references
listing for
author and year
method, the
literature is
alphabetically
arranged using
the first
author's name.

Disclaimer

This document was developed specifically for IT/SIWES students to support their report writing process. It is intended for educational purposes and should not be regarded as a universal or official standard as report formats may vary between universities, departments or the specific type of report being written.

This document uses insights from:

Akanbi C. T. and Oladebo K. T. (2024). Literature Citing and Listing: CVE 401-Technical Report Writing
Baze University Abuja Report Writing Template.

Gerson, S. J., & Gerson, S. M. (2020). *Technical Communication: Process and Product* (9th ed.). Pearson Education.

Kenig, E. Y., Gorak, A., Schoenmakers, H., & Kucka, L. (2001). Reactive and non-reactive distillation: Modelling and simulation. *Chemical Engineering and Processing: Process Intensification*, 40(3), 293–311. [https://doi.org/10.1016/S0255-2701\(00\)00150-8](https://doi.org/10.1016/S0255-2701(00)00150-8)

Kister, H. Z. (1992). *Distillation design*. McGraw-Hill Education.